

Mehak Panda

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SUMMARY

AI/ML-focused Computer Science undergraduate with experience in deep learning, machine learning, and research applications in healthcare and power systems. Skilled in PyTorch, Scikit-learn, and large-scale data analysis.

EDUCATION

National Institute of Science and Technology Berhampur, India B.Tech in Computer Science and Engineering
Sept 2023 – 2027

SKILLS

- **Languages:** Python, Java, SQL
- **Frameworks/Tools:** PyTorch, Streamlit, Scikit-learn
- **Libraries:** NumPy, Pandas, Matplotlib, Seaborn
- **Domains:** Machine Learning, Deep Learning, DSA

EXPERIENCE

1. Summer Research Intern May 2026 – Present
National Institute of Technology Jamshedpur

- Conducting research in Artificial Intelligence and Deep Learning for power system applications.
- Reviewing and analyzing research papers related to fault diagnosis and fault location estimation.
- Collaborating with faculty members to design and evaluate AI-driven solutions.
- Developing deep learning models and documenting research findings.

2. Summer Research Intern Jun 2026 – Present
National Institute of Technology Silchar

- Conducting research in healthcare-focused deep learning applications.
- Studying ECG signal processing and arrhythmia classification techniques.
- Performing data preprocessing, model training, and performance evaluation.
- Preparing technical documentation and experimental reports.

PROJECTS

1. AI-Based Digital Protection Relay for Transmission Line Fault Diagnosis and Location Estimation

- Developed a CNN-BiLSTM model for transmission line fault diagnosis and fault location estimation.
- Utilized an IEEE DataPort dataset containing over 579,000 records.
- Implemented data preprocessing, model training, and model evaluation using PyTorch.
- Conducted comparative analysis based on insights from research literature.

2. Automated Arrhythmia Detection from Single-Lead ECG Signals using Deep Convolutional Neural Networks

- Developed a 1D CNN model for automated ECG heartbeat classification.
- Trained and evaluated the model using the MIT-BIH Arrhythmia dataset.
- Classified heartbeats into five AAMI-recommended arrhythmia categories.
- Applied preprocessing techniques and addressed class imbalance challenges.

PUBLICATIONS

1. Revolutionizing Education with Remote Experimentation and Learning Analytics

- Published in IGI Global (March 2025).

2. Automation in the Agriculture Sector using AI/ML

- Accepted for publication, Tanz Research Journal (2026).

CERTIFICATIONS & ACHIEVEMENTS

- IBM SkillsBuild – Getting Started with Artificial Intelligence (2026)
- Data Analytics and Visualization Job Simulation (2024)
- Hackathon Finalist – Hack for Tomorrow, VSSUT Burla (2025)